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Bisphenol S instead of Bisphenol A: Toxicokinetic investigations in the ovine materno-feto-placental unit

Flore C. Grandin[a], Marlène Z. Lacroix[c], Véronique Gayrard[a], Glenn Gauderat[a], Hanna Mila[a], Pierre-Louis Toutain[b], Nicole Picard-Hagen[a,*]

[a] Toxalim, INRA (Institut National de la Recherche Agronomique), Université de Toulouse, ENVT (Ecole Nationale Vétérinaire de Toulouse), EIP (Ecole d'Ingénieurs de Purpan), UPS (Université Paul Sabatier), Toulouse, France

[b] The Royal Veterinary College, University of London, London, United Kingdom

[c] INTHERES, Université de Toulouse, INRA, ENVT, Toulouse, France

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ABSTRACT

Bisphenol S (BPS) is widely used as a substitute for Bisphenol A in consumer products. Despite its potential endocrine-disrupting effects and widespread exposure, toxicokinetic data, particularly during the critical period of pregnancy, are not available for BPS. The objectives of our study were to evaluate the mechanisms determining fetal exposure to BPS and to BPS glucuronide (BPSG) and to compare them with those prevailing for BPA.

The disposition of BPS and BPSG was evaluated in the materno-fetal unit of the catheterized pregnant ewe model, following intravenous administrations of BPS and BPSG to mothers and their fetuses. In a second experiment, the rate of BPS accumulation in the fetal compartment was determined under steady-state conditions after repeated intravenous BPS administrations to the mother.

In the maternal compartment, BPS was mainly metabolized into BPSG and totally eliminated in urine. Only 0.40% of the maternal dose was transferred to the fetus. However, once in the fetal compartment, 26% of the fetal dose was rapidly eliminated through placental transfer, while 46% of BPS was metabolized into BPSG which remained trapped in the fetal compartment. Thus, the elimination of BPSG from the fetal compartment required its back-conversion into bioactive BPS, leading to an 87% enhancement of the fetal BPS exposure.

Our findings demonstrate that, despite the low materno-fetal placental transfer of BPS, this substitute for BPA is able to accumulate in the fetal compartment after repeated maternal exposure, leading to chronic fetal exposure to BPS in a range of concentrations similar to those of BPA.

1. Introduction

Owing to regulatory actions and serious concerns regarding widespread human exposure to Bisphenol A (BPA) and its adverse effects on human health, BPA has been replaced with structurally similar compounds such as Bisphenol S (BPS) (ANSES, 2013; USEPA, 2014). Due to its higher thermal stability, BPS is increasingly being used as an intermediate for the production of polycarbonate plastics and epoxy resins in a range of consumer products including canned foodstuffs, thermal paper and personal care products (Wu et al., 2018).

BPS is now ubiquitous in the environment and is found worldwide. As for BPA, human exposure to BPS may occur mainly through ingestion, but also by inhalation and transcutaneous routes (Wu et al., 2018). The estimated median daily dietary intake of BPS is 9.55 ng/kg body weight/day (Liao and Kannan, 2014). In humans, BPS is mainly metabolized in the liver into bisphenol S glucuronide (BPSG) and eliminated in urine (Gramec Skledar and Peterlin Mašič, 2016; Le Fol et al., 2015).

Human biomonitoring studies recently conducted in the general population of the USA and seven Asian countries revealed the presence of BPS in 81% of the 315 urine samples, consistent with its widespread exposure (Chen et al., 2016; Liao et al., 2012; Wu et al., 2018). Furthermore, the urinary concentrations as well as the detection rate of BPS in adults was reported to have increased from 2000 to 2014 in the

Abbreviations: BPS, Bisphenol S; BPA, Bisphenol A; BPSG, Bisphenol S glucuronide; BPAG, Bisphenol A glucuronide; TK, Toxicokinetic; LLOQ, lower limit of quantification; AUC, Area under curve; MRT, Mean residence time; $t_{1/2}$, half-time; Cl, clearance; V_{ss}, steady state volume of distribution

* Corresponding author at: UMR1331 Toxalim, Ecole Nationale Vétérinaire de Toulouse, 23 chemin des Capelles, BP 87614, 31076 Toulouse Cedex 3, France.
E-mail addresses: flore.grandin@inra.fr (F.C. Grandin), m.lacroix@envt.fr (M.Z. Lacroix), v.gayrard@envt.fr (V. Gayrard), h.mila@envt.fr (H. Mila), n.hagen-picard@envt.fr (N. Picard-Hagen).

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U.S., reflecting the increasing replacement of BPA with BPS (Ye et al., 2015).

More recently, BPS was found in four maternal and seven cord serum samples collected from China at low concentrations ranging from < 0.03 to 0.12 ng/mL (Liu et al., 2017), thus demonstrating a placental transfer of BPS from the mother to the fetus.

Due to the structural similarity of BPS with BPA and its widespread use, concerns about the safety of BPS have become an important issue. Some limited *in vitro* and *in vivo* data suggest that BPS may have endocrine disrupting properties similar to BPA (Eladak et al., 2015; Goldinger et al., 2015; Rochester and Bolden, 2015; Roelofs et al., 2015; Rosenmai et al., 2014). Furthermore, recent toxicological investigations in zebrafish, rodents and sheep have demonstrated that exposure to BPS during development can disrupt reproductive, metabolic and developmental endpoints through the endocrine pathways (Hill et al., 2017; Ji et al., 2013; Naderi et al., 2014; Pu et al., 2017; Qiu et al., 2016) indicating the vulnerability of the gestation period to BPS exposure.

The reliance on findings from animal studies to determine the human health consequences of BPS exposure during prenatal life emphasizes the importance of toxicokinetic (TK) approaches in animal models for predicting human fetal BPS exposure. The factors determining fetal exposure provide key information to determine if BPS is an acceptable alternative to BPA. The pregnant ewe is an acknowledged model frequently used to characterize drug disposition in the materno-feto-placental unit (Corbel et al., 2013; Kumar et al., 1997; Szeto, 1982; Yoo et al., 1993).

The aim of our study was to evaluate the toxicokinetic parameters determining the disposition of BPS and its main metabolite, BPSG, in the materno-feto-placental unit in the model of pregnant ewe and to compare them with those of BPA. The levels of BPSG sustained in the fetal compartment after a single maternal BPS administration in the first experiment prompted us to evaluate the rate of BPS accumulation in the fetal compartment after repeated maternal administrations.

2. Materials and methods

2.1. Animals

All animal procedures were carried out in accordance with accepted standards of humane animal care under agreement number 31-2011-142 from the French Ministry of Agriculture. The protocol was approved by the regional ethical committee (Midi-Pyrénées: protocol APAFiS #11899 - 2017102313532843).

The first experiment was carried out on eight pregnant Lacaune ewes with a mean (± SD) body weight of 79 ± 15 kg and the second experiment on 3 pregnant Lacaune ewes with a mean (± SD) body weight of 74 ± 3 kg. All animals received two commercial pelleted rations daily and hay and water were given *ad libitum*. For each experiment, one fetus per ewe was chronically catheterized between 109 and 113 days of gestation (pregnancy length 147 days) as previously described (Corbel et al., 2013).

2.2. Experimental design

2.2.1. Experiment 1: BPS and BPSG feto-maternal toxicokinetics

This first experiment was designed to evaluate the mechanisms determining fetal exposure to BPS and BPSG (Fig. 1).

During the first period, pregnant ewes received an intravenous (IV) bolus administration of non-labelled BPSG (6.3 μmol/kg, 2.7 mg/kg) and simultaneously, the catheterized fetus received an IV administration of BPS-d8 (19.4 μmol, 5 mg). During the second period, three days apart, simultaneous IV administrations of non-labelled BPS (20.0 μmol/kg, 5 mg/kg) to the mother, and of BPSG-d8 (40.3 μmol, 17.5 mg) to the fetus were given. BPS and BPSG were administered in deuterated forms (BPS-d8 and BPSG-d8) to the fetus so that the BPS and BPSG in the maternal and fetal compartments, resulting from maternal or fetal administrations, could be distinguished. These doses were selected on the basis of a previous TK study performed in ewes (Grandin et al., 2017) to allow correct estimation of TK parameters in the materno-fetal unit.

During both periods, fetal arterial and maternal jugular blood samples and amniotic fluid samples were drawn before and at 7, 15, 30 min, 1, 2, 4, 6, 8, 10, 24, 34, 48, 58 and 72 h after IV administrations. Total maternal urine was collected over 24 h with collections at 0, 1, 3, 6, 10, 15, 20 and 24 h after BPS/BPS-d8 and BPSG/BPSG-d8 administrations.

2.2.2. Experiment 2: repeated BPS dosing in pregnant ewe

This experiment was designed to evaluate the rate of BPS accumulation in the fetal compartment resulting from repeated maternal BPS administrations (Fig. 1). The feto-maternal TK results obtained in the first experiment revealed the long persistence of BPS in the fetal compartment, determined from the rate of BPSG hydrolysis, which indicated that for both maternal and fetal BPS plasma concentrations to rapidly reach steady-state, an initial loading dose of BPS was required. A rough simulation of the time course of fetal concentrations led to selection of the following dosing protocol with a loading dose of BPS (75 mg/kg) infused for 1 day at a flow rate of 2.3 mL/h to the pregnant ewe, followed by seven daily IV boluses of BPS maintenance doses (5 mg/kg).

Fetal and maternal blood samples were drawn before and at 5, 15, 30 min, 1, 2, 3, 4, 6, 8, 10 h after the BPS infusion, and every 15 min during the last hour of the infusion to determine the steady-state concentrations. Fetal and maternal blood samples were then drawn daily just before BPS administrations and at 5, 15, 30 min, 1, 2, 3, 4, 6, 8, 10 and 24 h after the 7th IV bolus administration.

2.3. Solution preparation and animal treatments

All materials for the preparation of solutions, including materials used for sampling, processing and analysis, were in glass or Bisphenols-free plastic (polypropylene). BPS (purity ≥98%) was purchased from Sigma-Aldrich (Saint-Quentin Fallavier, France). BPS-d8 (purity ≥98%) and BPSG-d8 (purity ≥97%) were purchased from Toronto Research Chemicals (Toronto, Canada). For experiment 1 and bolus administrations in experiment 2, BPS and BPS-d8 were extemporaneously dissolved in ethanol/ultrapure water/propylene glycol (10/11/79, vol/vol/vol) at concentrations of 50 and 5 mg/mL, respectively. For infusion in experiment 2, BPS was extemporaneously dissolved in ethanol/propylene glycol (1/49, vol/vol) at concentration of 100 mg/mL.

For experiment 1, fractions of BPSG purified (68–85%) as previously described (Grandin et al., 2017) were used for maternal administrations. These fractions contained a small amount of BPS (< 4.4%). BPSG and BPSG-d8 were extemporaneously dissolved in 0.9% NaCl buffer at concentrations of 50 and 10 mg/mL, respectively.

For maternal administrations, the volumes of BPS and BPSG solutions were adjusted to the body weights recorded on the three previous days. Maternal intravenous doses were administered into the left jugular vein *via* an indwelling catheter and fetal intravenous doses were administered *via* a catheter chronically implanted in the jugular vein. For intravenous infusions, the catheter was connected to a portable syringe pump (Graseby® MS 32), as previously described (Collet et al., 2010).

2.4. Sample collection

Maternal blood samples (3 mL) were obtained by direct puncture of the right jugular vein. Fetal blood samples (0.7 mL) were collected *via* an indwelling catheter chronically implanted in the fetal carotid. Amniotic fluid samples (1 mL) were drawn *via* an indwelling catheter permanently implanted in the amniotic sac. Total urine was collected

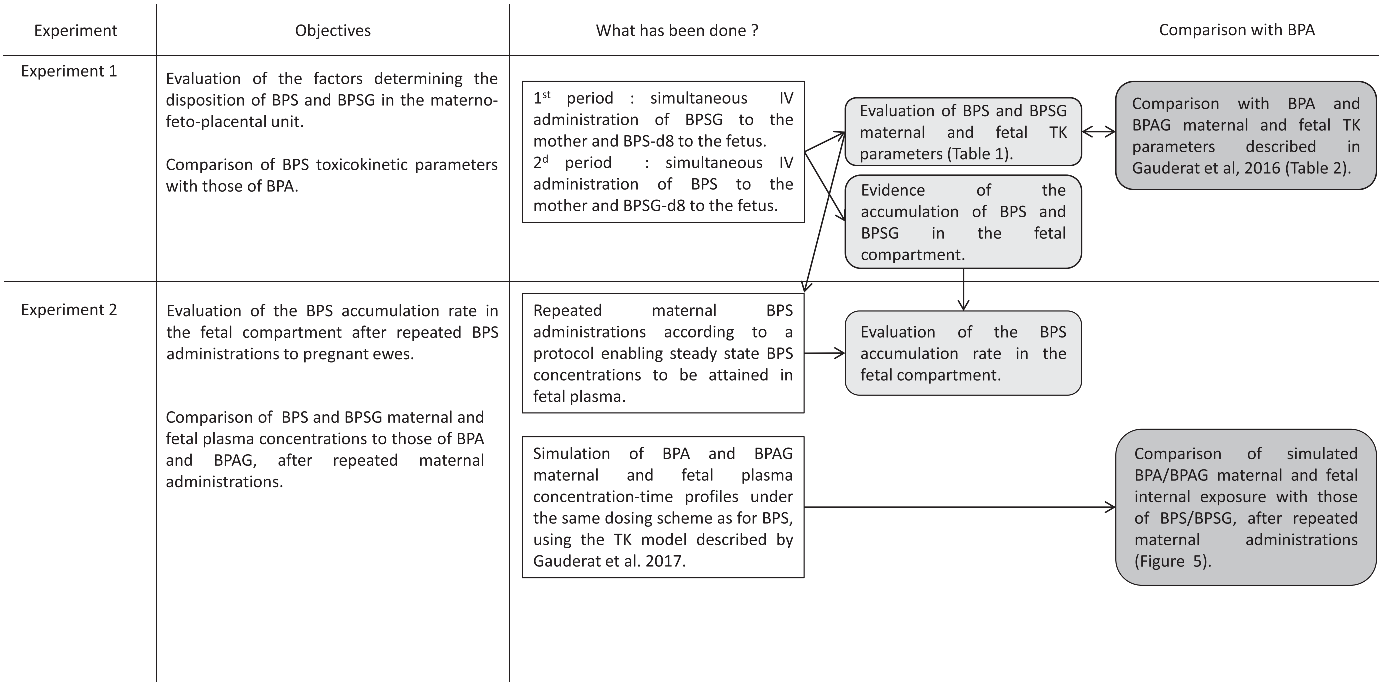

Fig. 1. Workflow diagram of the main steps in the toxicokinetic investigations to evaluate the mechanisms determining the disposition of BPS and its main metabolite in the materno-fetal unit.

via a Foley catheter aseptically placed in the urethra of the ewes.

Blood samples were collected in heparinized tubes, immediately chilled in ice and centrifuged for 10 min at 3000*g* and 4 °C. Plasmas were separated and stored at −20 °C until assayed.

Amniotic fluid and urine samples were collected in polypropylene tubes, immediately chilled in ice, centrifuged for 10 min at 3000*g* and 4 °C and the supernatants were stored at −20 °C until assayed.

2.5. Sample analysis

The BPS, BPS-d8, BPSG and BPSG-d8 in the samples were simultaneously quantified by on-line solid phase extraction with ultra-performance liquid chromatography coupled to tandem mass spectrometry (Acquity-2D UPLC® Xevo® TQ, Waters, Milford, MA, USA) according to a method validated in compliance with European Medicines Agency guidelines (Grandin et al., 2017). This method was adapted and validated for urine samples and for amniotic fluid samples. Briefly, 100 μL of plasma or amniotic fluid samples were precipitated with 100 μL of acetonitrile/zinc sulphate containing internal standard BPS-d4 (100 ng/mL). Urine samples (100 μL) were diluted with 200 μL of the above-mentioned mixture containing internal standard at 1000 ng/mL. Both BPS and its metabolite were extracted with an on-line C8 SPE cartridge and separated on a CSH C18 column with water/acetonitrile gradient elution. The multiple reaction monitoring transitions used to quantify BPS, BPS-d8, BPSG and BPSG-d8 were 249 > 108, 257 > 112, 425 > 249, 433 > 257, respectively.

Blank samples were used to check the absence of contamination during assays. The accuracy and the intra- and inter-day precisions of the method were evaluated from quality control samples at three concentration levels.

The precisions, expressed by the coefficients of variation were below 15% at each concentration level and for the four analytes. The lower limits of quantification (LLOQ) were validated at 3 ng/mL for BPS and 1 ng/mL for BPS-d8 in plasma and at 1 ng/mL for both analytes in amniotic fluid. The LLOQ were validated in urine at 100 ng/mL for BPS and 50 ng/mL for BPS-d8. For BPSG and BPSG-d8, LLOQ were validated at 10 ng/mL in plasma and amniotic fluid and at 100 ng/mL in urine.

2.6. Toxicokinetic analysis

The plasma concentrations-times profiles of BPS, BPS-d8, BPSG and BPSG-d8 observed in the first experiment were analyzed with a sparse-data non-compartmental approach using Phoenix® NLME™ (version 6.4; Pharsight, Mountain view, CA, USA). The sparse data approach computes the mean concentration *versus* time curve of data, based on the mean concentration value for each nominal time value (Gabrielsson and Weiner, 2012; Nedelman et al., 1995). All plasma, amniotic fluid and urine concentrations were converted into molar concentrations.

In the first experiment, the area under the plasma concentration time-curve (AUC_{last}) from dosing time to the last sampling time was calculated using the trapezoidal linear/log interpolation rule. The AUC from t = 0 to infinity was obtained by adding to AUC_{last}, the area extrapolated from the last observation to infinity by dividing the last observed quantifiable plasma concentration by the slope of the terminal phase as estimated by linear regression, using the best fit option of Phoenix®.

The clearance (Cl), mean residence time (MRT), terminal half-life ($t_{1/2}$) and the steady state volume of distribution (V_{ss}) of BPS (BPS-d8) and BPSG (BPSG-d8) were computed using classical pharmacokinetic equations described in Supplementary material (Gibaldi and Perrier, 1982). Because of the extent of BPSG deconjugation-reconjugation recycling, which led to an enhancement of BPS fetal exposure, the BPS fetal TK parameters were computed over 10 h post-administration, to estimate the first slope of elimination of BPS from the fetal compartment.

For each mother (fetus), the extent of BPS (BPS-d8) glucuronoconjugation over the sampling period was calculated by multiplying the maternal BPSG (fetal BPSG-d8) value of AUC from t = 0 to infinity obtained after IV maternal BPS (fetal BPS-d8) administration by the corresponding maternal (fetal) clearance of BPSG (BPSG-d8) (see Supplementary material).

The extent of BPSG-d8 hydrolysis over the sampling period was calculated by multiplying the BPS-d8 AUC_{last} value obtained after BPSG-d8 administration by the corresponding fetal BPS-d8 clearance (see Supplementary material).

The maternal to fetal transfer of BPS for each mother-fetus pair over the sampling period was estimated from the product of the fetal BPS value of AUC from t = 0 to infinity obtained after IV maternal BPS administration and the corresponding BPS-d8 fetal clearance divided by the IV maternal BPS dose administered (see Supplementary material).

The fetal to maternal transfer of BPS-d8 for each mother-fetus pair over the sampling period was estimated from the product of the maternal BPS-d8 value of AUC from t = 0 to infinity obtained after IV fetal BPS-d8 administration and the corresponding BPS maternal clearance divided by the IV fetal BPS-d8 dose administered (see Supplementary material).

The degree of Exposure Enhancement of BPS (EE_{BPS}) and BPSG (EE_{BPSG}), afforded by reversible metabolism of BPSG into BPS was calculated as follows (Cheng and Jusko, 1993):

$$EE_{BPS} = EE_{BPSG} = 1 + \frac{Cl_{fcBPS} \times Cl_{fBPSG}}{(Cl_{ftBPS} + Cl_{fcBPS}) \times Cl_{fBPSG} - Cl_{fcBPS} \times Cl_{fBPSG}} \quad (1)$$

where Cl_{fcBPS} is the fetal BPS glucuronoconjugation clearance, Cl_{fBPSG} is the fetal clearance of BPSG, and Cl_{ftBPS} is the fetal to maternal BPS clearance.

In experiment 2, the plasma concentrations-times profiles of BPS and BPSG were analyzed by using a sparse data non-compartmental approach. The terminal half-life of BPS and of BPSG was computed using classical pharmacokinetic equations (Gibaldi and Perrier, 1982). Assuming that BPS TK is linear, the BPS accumulation in the fetal compartment was calculated for each ewe as the ratio of the AUC_{last} under steady-state conditions (7th bolus administration) divided by the AUC_{last} after the 24 h-infusion, normalized by doses (Toutain and Bousquet-Mélou, 2004) (see Supplementary material).

3. Results

The concentrations of BPS, BPS-d8, BPSG and BPSG-d8 in all the control samples drawn before the first dosing were systematically below the LLOQ of the analytical method, thereby indicating that no sample contamination had occurred during sample collection, processing and assay.

3.1. Experiment 1: BPS and BPSG feto-maternal toxicokinetics

The experiment was performed on 7 fetuses (4 females 3 males) with a mean (± SD) body weight of 2.5 ± 0.4 kg. The average age of fetuses was 117 ± 0.5 gestation days at the beginning of the experiment.

Only the quantities of BPSG in maternal urine are reported since BPS represented < 0.5% of the total BPS recovered in urine. Cumulative amounts of BPSG excreted in maternal urine were successfully estimated after BPSG and BPS administrations to pregnant ewes for 8 ewes and 4 ewes out of 8, respectively.

3.1.1. Simultaneous fetal BPS-d8 and maternal BPSG IV administrations

Fig. 2 shows the time course of mean (± SD) BPS-d8, BPSG and BPSG-d8 concentrations in maternal (A) and fetal (B) plasma and in amniotic fluid (D) and the cumulative amounts of BPSG excreted in the urine (C) of pregnant ewes following simultaneous fetal BPS-d8 (19.4 µmol, 5 mg) and maternal BPSG (6.3 µmol/kg, 2.7 mg/kg) dosing (see Supplementary data, Tables S1, S3, S5 and S7).

After BPSG administration to adult ewes, the maternal plasma BPSG concentrations decreased to reach values close to the LLOQ about 34 h after BPSG administration. No BPS was detected in maternal plasma. Moreover, no BPSG was detected in fetal plasma or in amniotic fluid after BPSG maternal administration. The maximum BPSG transfer in the materno-fetal direction, estimated by multiplying the LLOQ of the BPSG assay (10 ng/mL) by the fetal BPSG clearance, was < 0.0001%.

Almost all the BPSG dose administered to the mother (94 ± 25%) was recovered in urine during the 24 h following BPSG administration, mostly (85 ± 8%) within the first 3 h after BPSG dosing (Fig. 2C).

After fetal BPS-d8 administration, the BPS-d8 fetal plasma concentrations decreased rapidly during 24 h post-administration and then tended to remain relatively constant. The fetal plasma BPSG-d8 concentrations, that attained maximum concentrations 4 h post-administration, decreased very slowly over the sampling period (Fig. 2B). In maternal plasma, the BPS-d8 concentrations that were maximal at 15 min post-fetal administration, decreased rapidly according to a single log linear process. Maternal BPSG-d8 concentrations peaked 1 h after BPS-d8 dosing to the fetus, after which the BPSG-d8 concentration-time profile showed biphasic decay, the BPSG-d8 concentrations reaching values below the LLOQ 10 h post-dosing (Fig. 2A).

In the amniotic fluid, BPS-d8 concentrations showed a transitory increase, reaching maximum values 2 h after the administration and then remained relatively constant and close to the plasma levels from 10 h post-dosing and throughout the sampling period. BPSG-d8 concentrations in amniotic fluid increased gradually to reach maximum values about 24 h post-dosing and then remained at a relatively constant level around ten times lower than the plasma levels over the sampling period (Fig. 2D).

3.1.2. Simultaneous fetal BPSG-d8 and maternal BPS IV administrations

Fig. 3 shows the time course of mean (± SD) BPS, BPS-d8, BPSG and BPSG-d8 concentrations in maternal (A) and fetal (B) plasma and amniotic fluid (D) and the cumulative amounts of BPSG excreted in urine (C) of pregnant ewes following simultaneous IV administrations of maternal BPS (20.0 µmol/kg, 5 mg/kg) and fetal BPSG-d8 (40.3 µmol, 17.5 mg) (see Supplementary data, Tables S2, S4, S6 and S8).

After BPS administration to adult ewes, maternal BPS concentrations decreased rapidly and reached values below the LLOQ 10 h post-administration (Fig. 3A). Maternal plasma BPSG concentrations that were maximal at 15 min post-BPS administration decreased rapidly to reach values below the LLOQ 72 h after BPS administration. Ninety (± 34) percent of the BPS dose administered to pregnant ewes was excreted in the maternal urine during the first 24 h post-dosing, mostly (68 ± 22%) during the first 3 h after BPS administration (Fig. 3C).

In the fetal compartment, BPS plasma concentrations peaked at the first sampling time (7 min), decreased rapidly until 10 h post-administration and then remained relatively steady over the sampling period. BPSG concentrations, resulting from fetal metabolism, reached maximum concentrations about 6 h post-dosing and then decreased very slowly over the sampling period (Fig. 3B).

In the amniotic fluid, both BPS and BPSG levels increased during the few hours post-dosing and then remained relatively constant over the sampling period (Fig. 3D).

After BPSG-d8 administration to the fetus, the slow decrease in fetal plasma BPSG-d8 concentrations was associated with low and relatively steady BPS-d8 concentrations (Fig. 3B). BPSG-d8 was detected in the maternal urine but at levels below the LLOQ (100 ng/mL for BPSG-d8), which means that < 0.6% of the fetal BPSG-d8 dose was transferred from the fetus to its mother during the first 24 h.

In the amniotic fluid, BPS-d8 and BPSG-d8 concentrations increased progressively during the few hours post-dosing and then remained relatively constant throughout the sampling period (Fig. 3D). The BPSG-d8 levels in amniotic fluid were constantly lower than those of plasma, whereas the BPS-d8 concentrations in plasma and amniotic fluid were similar.

Interestingly, the ratios of BPSG/BPS (BPSG-d8/BPS-d8) during the terminal phase were higher in fetal plasma (about 200 times) than in the amniotic fluid (about 25 times), whether after BPS-d8 or BPSG-d8 fetal administration or after BPS maternal administration.

3.2. Toxicokinetic parameters

The BPS and BPSG TK parameters from ovine fetuses and adult ewes

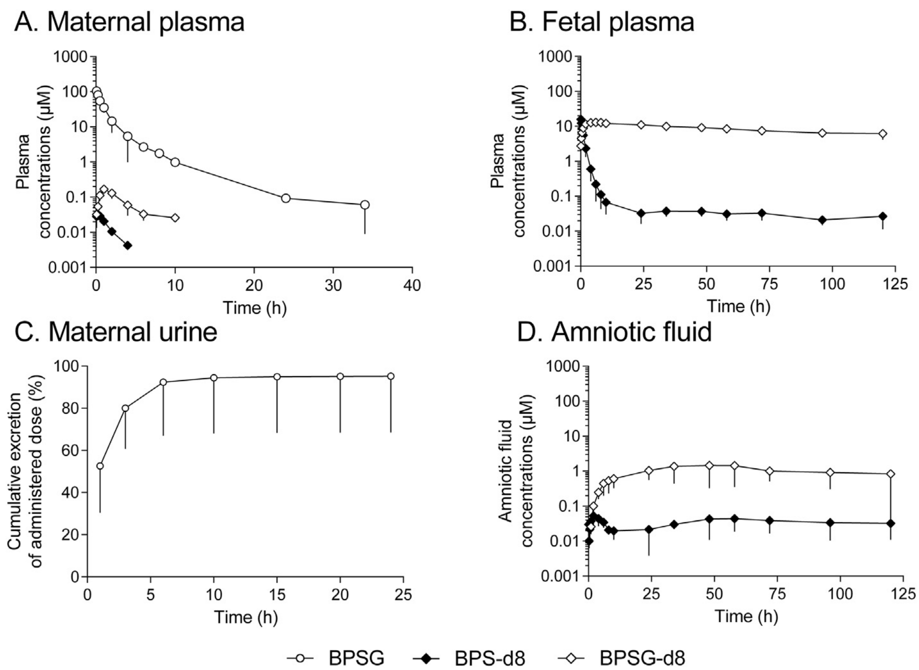

Fig. 2. Semilogarithmic plots of mean (± SD) concentration-time-profiles (μM) for BPS-d8 (closed diamonds), BPSG (open circles) and BPSG-d8 (open diamonds) in maternal (A), fetal (B) plasma and amniotic fluid (C) and for BPSG in maternal urine (D) following simultaneous IV administrations of BPSG (6.3 μmol/kg, 2.7 mg/kg, n = 8) to the mother and BPS-d8 (19.4 μmol, 5 mg, n = 7) to the fetus.

are summarized in Table 1. Among the BPS TK parameters measured in the mother-fetus pairs, BPS plasma clearance was about two times less in the fetus than in its mother. Fetal BPS half-life, which is influenced by both plasma clearance and volume of distribution, was similar to maternal BPS half-life (1.92 h and 2.37 h). Among the BPSG TK parameters measured in the mother-fetus pairs, BPSG plasma clearance was 25 times lower in the fetus than in the pregnant ewe. The BPSG half-life in the fetal compartment was 18 times higher than the maternal one (93.6 h *versus* 5.22 h).

The computation of TK parameters and the BPS and BPSG concentrations profiles described in the mother-fetus pairs in experiment 1 allowed evaluation of the transfer of BPS/BPSG from one compartment to the other. The estimated mean fraction of BPS transferred from the mother to the fetus was 0.40%. The mean fraction of BPS-d8 transferred from the fetus to the mother, estimated from the time course of plasma

Table 1
Mean toxicokinetic parameters of BPS and BPSG from ovine fetuses and pregnant sheep administered a single IV dose of BPS (BPS-d8) and BPSG (BPSG-d8).

Toxicokinetic parameters	BPS		BPSG	
	Fetus (n = 8)	Mother (n = 6)	Fetus (n = 4)	Mother (n = 9)
Cl (mL / (h ∗ kg))	423	993	1.87	48.6
Vss (mL/kg)	597	380	251	121
MRT (h)	1.41	0.383	134	2.49
$t_{1/2}$ (h)	1.92	2.37	93.6	5.22

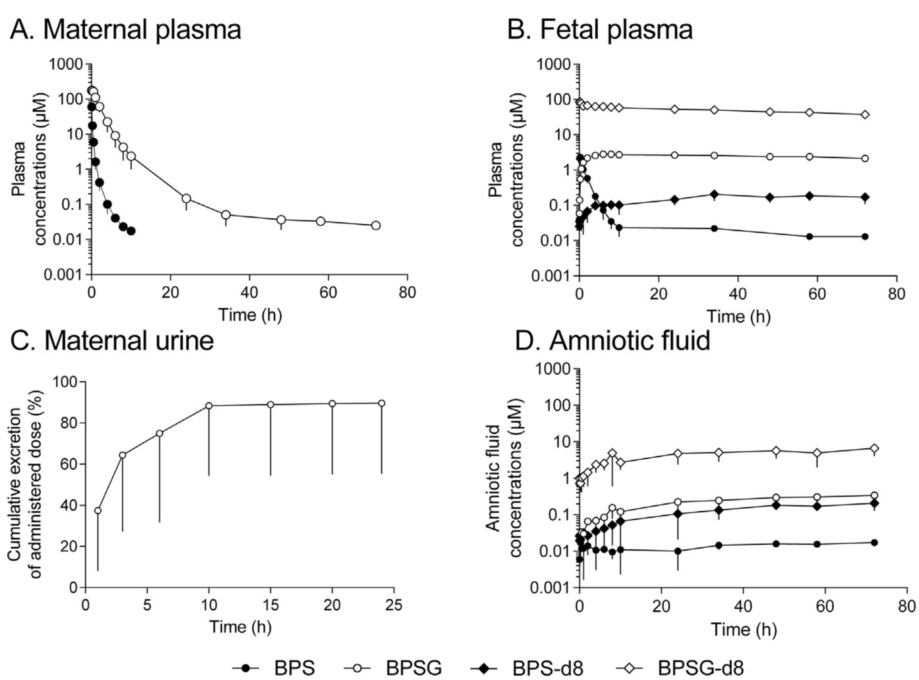

Fig. 3. Semilogarithmic plots of mean (± SD) concentration-time-profile (μM) for BPS (closed circles), BPS-d8 (closed diamonds), BPSG (open circles) and BPSG-d8 (open diamonds) in maternal (A), fetal (B) plasma and amniotic fluid (C) and for BPSG in maternal urine (D) following simultaneous IV administrations of BPS (20.0 μmol/kg, 5 mg/kg, n = 6) to the mother and of BPSG-d8 to the fetus (40.3 μmol 17.5 mg, n = 4).

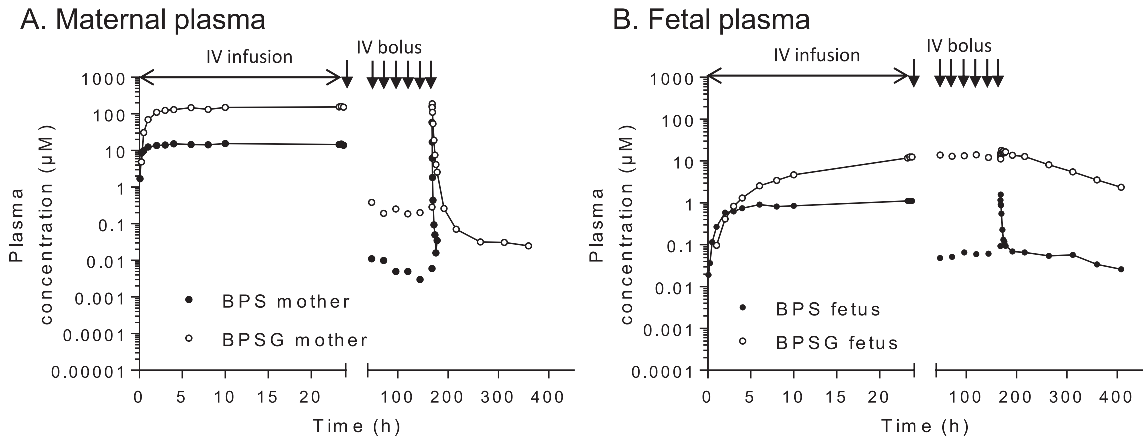

Fig. 4. Semilogarithmic plots of representative concentration-time-profiles (μM) for BPS (closed circles) and BPSG (open circles) in maternal (A) and fetal (B) plasma after an IV infusion loading dose of BPS (300 μmol/kg, 75 mg/kg) followed by the administration of 6 repeated BPS IV bolus maintenance doses (20 μmol/kg, 5 mg/kg) to the mother.

maternal BPS-d8 concentrations, was 26.3% over the 72 h sampling period.

The estimated fraction of BPS metabolized into BPSG in the maternal compartment was 96.7%. The fraction of BPS-d8 metabolized into BPSG-d8 in the fetal compartment calculated over the period of 72 h, was 24.9% but extrapolation of the AUC to infinity indicated that this fraction might attain 46.4%. The estimated fraction of BPSG-d8 hydrolysed into BPS-d8 in the fetal compartment was only 29.5% of the fetal dose over the sampling period of 72 h and reached 100% when the BPS-d8 AUC was extrapolated to infinity.

The degree of exposure enhancement of BPS and of BPSG due to recycling *per se*, estimated from the BPS and BPSG fetal clearances was 1.87.

3.3. Experiment 2: repeated BPS administrations to pregnant ewes

This experiment was performed on one female and 2 male fetuses with a mean (± SD) body weight of 3.5 ± 0.8 kg. The average age of the fetuses was 117 days at the beginning of the experiment. Fig. 4 shows a representative time course of maternal and fetal plasma BPS and BPSG concentrations after the administration of an IV infusion loading dose of BPS (75 mg/kg) followed by seven daily maintenance IV bolus doses (5 mg/kg) of BPS to pregnant ewes (see Supplementary data, Tables S9 and S10). The steady state conditions, evaluated from the intra-individual coefficients of variation of the trough BPS and BPSG fetal concentrations before the 6 daily maintenance IV bolus doses, were 20.7% for BPS and 4.7% for BPSG.

After a BPS loading dose administration to pregnant ewes, the fetal and maternal BPS and the maternal BPSG plasma concentration-time profiles were characterized by a plateau attained 6 h after the beginning of the BPS infusion, whereas the BPSG fetal plasma concentrations continued to increase until the end of the maternal BPS infusion. The fetal concentrations of BPS and BPSG were about ten times lower than those in maternal plasma. By contrast, trough BPS and BPSG concentrations in fetal plasma, associated with repeated maternal exposure dosing with BPS, were much higher than the corresponding pattern obtained for maternal plasma. The sustained fetal exposure to BPSG was reflected by the high ratio of around 75 between the basal fetal and maternal concentrations of BPSG at steady-state.

After the last BPS maternal administration, maternal BPS plasma concentrations dropped rapidly to values below the LLOQ 24 h post-dosing. Maternal BPSG and fetal BPS concentrations displayed biphasic decay, with a rapid decrease during the 10 h post-dosing. Thereafter, the maternal BPSG and fetal BPS and BPSG concentrations decreased slowly with terminal half-lives of the same order of magnitude (137 h, 118 h and 98.5 h, respectively), (Fig. 4A and B). The individually calculated accumulation ratios for BPS in the fetal compartment following repeated maternal BPS administrations were 5.82, 3.42 and 10.2.

4. Discussion

Pregnancy represents a period of vulnerability to BPS exposure (Catanese and Vandenberg, 2017; Crump et al., 2016). Human exposure to BPS, particularly during fetal development is to date unknown. This highlights the urgent need to characterize the disposition of BPS and its main metabolite (BPSG) in the feto-maternal unit. The present study was based on concurrent determination of BPS and BPSG (or deuterated forms) TK in a maternal and late-stage fetal ovine model, relevant to human gestation. So far, this study is the first to report the factors controlling fetal exposure to BPS and BPSG and to compare BPS and BPSG fetal TK parameters with those of BPA and BPAG, previously described in the ovine model (Table 2, Gauderat et al., 2016).

We have shown that only 0.40% of the maternal BPS dose was transferred to the fetus, which means that the fetus was exposed to a dose related to its body weight, that was ten times lower than the maternal BPS dose. However, once in the fetal compartment, only 26% of the BPS dose was rapidly eliminated through placental transfer, while at least 46% was glucurono-conjugated by the fetus. The very slow elimination of BPSG from the fetal compartment due to deconjugation-reconjugation recycling and to the restricted placental passage

Table 2
Mean (± SD) toxicokinetic parameters of BPA and BPAG in the ovine fetus and pregnant ewe (Gauderat et al., 2016).

Toxicokinetic parameters	BPA		BPAG	
	Fetus (n = 8)	Mother (n = 8)	Fetus (n = 8)	Mother (n = 7)
Cl (mL / (h * kg))	14,840 ± 4071	1746 ± 380	20.4 ± 12.8	334 ± 76.0
Vss (mL/kg)	5090 ± 1424	1281 ± 302	701 ± 218	324 ± 95.7
MRT (h)	0.343 ± 0.0261	0.746 ± 0.148	39.3 ± 10.9	0.842 ± 0.543
$t_{1/2}$ (h)	0.307 ± 0.0302	1.55 ± 0.462	28.3 ± 6.41	3.17 ± 1.84

of BPS, leads to an accumulation of BPS in the fetal circulation. Indeed, BPS accumulated as BPSG in the fetal compartment has to be deconjugated into its active form before it can be ultimately eliminated by the placenta.

After maternal BPS administration, almost all the BPS dose entering maternal circulation was extensively glucurono-conjugated and rapidly eliminated in urine, *via* a non-placental route. This result is consistent with TK data obtained in mice (Song et al., 2018). Pregnancy did not modify the TK parameters of BPS (Grandin et al., 2017), which can be explained at least in part by the low placental BPS transfer from the mother to its fetus. The maternal BPS and BPSG clearances were twice and seven times lower than those of BPA and BPAG, respectively, indicating that the capacity of the mother to eliminate BPS and BPSG was lower than for BPA (Table 2, (Gauderat et al., 2016)). Despite the structural similarity between these two bisphenols, the lower rate of BPS glucuronidation compared to that of BPA could be attributed to changes in molecular properties that affected their interactions with metabolic enzymes. Indeed, different UDP glucuronosyltransferases enzymes are involved in the glucuronidation of BPA and BPS (Gramec Skledar et al., 2014). The three times lower steady-state distribution volumes of BPS and BPSG compared to those of BPA and BPAG were consistent with their lower lipophilicities (LogP BPS = 1.65, LogP BPSG = −0.33 *versus* LogP BPA = 3.43, LogP BPAG = 1.12). This weak tissular distribution has already been reported in mice (Song et al., 2018).

BPS entering into the fetal circulation was slowly eliminated with a forty times lower fetal clearance than for BPA, resulting from the lower feto-maternal placental clearance (about 100 times lower) and glucuronidation clearance (about twenty times lower). These data support the assumption that bisphenol S and A are glucuronidated by different isoforms of UDP glucuronosyltransferases enzymes which could differ in the development of their expression during pregnancy (Gramec Skledar et al., 2014). The placental transfer of BPS from fetus to mother appeared to be extremely rapid with maximum BPS concentrations occurring in maternal plasma within the first 7 min after a fetal IV bolus administration. However, it was estimated that only 26% of the total fetal dose was eliminated by placental transfer during the first hours following a fetal BPS administration. By contrast, the BPS dose fraction glucuronidated by the fetus (46%) was eliminated very slowly, resulting in a much higher fetal exposure to BPSG than to BPS (ratio of about 200) due to the much lower fetal BPSG clearance. In our experiments, we only measured the glucuronidated form of BPS, and not BPS sulphate. Indeed, *in vitro* studies on human hepatocarcinoma cell line (Le Fol et al., 2015) or with human liver and intestine microsomes showed that glucuronidation is the predominant metabolic pathway for BPS in adults (Skledar et al., 2016). In addition, even if expression of liver phenol sulfoconjugation activities has already been reported in midgestation fetuses (Hines, 2008), we have previously demonstrated that the contribution of hepatic sulfoconjugation to BPA detoxification in the late pregnancy ovine fetus is low (Corbel et al., 2013).

The limited placental transfer of BPS, compared to BPA, both in materno-fetal (15 times lower) and in feto-maternal (three times lower) directions could be explained by its weak lipophilicity. But, considering the high doses of BPS used in our TK experiment, a saturation of the placental transfer cannot be ruled out. Furthermore, given the difference in placental structure between ovine and human (Barry and Anthony, 2008), with fewer barriers separating the human maternal and fetal blood compartments, the actual extent of BPS placental transport should be further investigated in a human placental perfusion model, as has already been evaluated for BPA (Corbel et al., 2014).

The fate of BPSG in the feto-maternal unit was evaluated after BPSG-d8 administration directly into the fetal circulation. The persistence of BPSG-d8 in the fetal compartment was long (terminal half-life of 93 h, three times higher than that of BPAG), associated with a low fetal BPSG clearance), ten times lower than that of fetal BPAG. The extremely limited BPSG placental transfer (< 0.0001%) in the materno-fetal direction was similar to those obtained for BPAG both in the sheep model and in the pregnant rat (Nishikawa et al., 2010). In addition, in the pregnant PBPK model in both rat and human (Kawamoto et al., 2007; Sharma et al., 2018), the diffusion coefficients of BPAG from the placenta to the fetus are low.

Moreover, using TK parameters, the estimated fraction of BPSG-d8 hydrolysed into BPS-d8 represented only 29.5% of the total fetal BPSG-d8 dose over the sampling period, and reached 100% when the BPS-d8 AUC was extrapolated to infinity. This suggests that almost all the BPSG has to be back-converted into its active form (BPS) in order to be ultimately eliminated by the placenta, which leads to the gradual re-entry of BPS into the fetal compartment. The degree of exposure enhancement to BPS in the fetal compartment was 1.87 (Cheng and Jusko, 1993), indicating that back-metabolism of BPSG to BPS increased the fetal exposure to BPS by 87%.

The repeated BPS maternal administration protocol (experiment 2) enabled us to correctly estimate the terminal half-lives of both BPS (including back-formed BPS) and BPSG to better characterize the BPS accumulation in the fetal compartment. The highly prolonged BPS $t_{1/2}$ in the fetal compartment (118 h), about 50 times higher than the BPS $t_{1/2}$ estimated from a single fetal BPS administration and similar to that of fetal BPSG $t_{1/2}$, indicates that the very slow elimination of BPS from the fetal compartment is controlled by the reversible metabolism of BPSG. This process leads, in turn, to a prolonged maternal BPSG elimination half-life (about 25 times higher than the elimination half-life of BPSG calculated from a single maternal BPSG administration) because of the constant re-entry and glucuronoconjugation, within the maternal compartment, of the estimated 26% of the dose of back-converted BPS. These long terminal half-lives of fetal back-formed BPS and BPSG result in the accumulation and trapping of BPS in the fetal compartment, the mean BPS accumulation ratio being 6.5 after repeated maternal BPS administrations once daily under equilibrium conditions. The context of repeated maternal exposure to BPS, due to the very slow back conversion of BPSG to BPS and to the limited placental transfer, leads to a specific pattern of BPS and BPSG exposure characterized by much higher trough BPS and BPSG concentrations in the fetal plasma, compared to those in the maternal plasma.

The comparison of BPS/BPSG terminal phases of the concentration profiles in plasma and in amniotic fluid, either after maternal BPS or fetal BPS/BPSG administrations showed that the BPS levels in amniotic fluid were of similar magnitude to the corresponding levels in plasma. By contrast, the levels of BPSG were around six times higher in plasma than in the amniotic fluid. These data indicate that BPS and its metabolite may be excreted in the amniotic fluid and that the fetus can be re-exposed to this excreted BPS/BPSG either by swallowing the amniotic fluid or through dermal absorption. The lower ratio of the BPSG/BPS levels in amniotic fluid compared to that in plasma (around 30 *versus* 200) might be due to the local action of β-glucuronidases regenerating BPS in the amniotic fluid, placenta or fetal kidney and by the recirculation of BPS/BPSG between the placenta and the amniotic sac (Ginsberg and Rice, 2009; Kamarýt and Matýsek, 1980). Thus, BPSG should be viewed as a sustained precursor for BPS supply to sensitive fetal target tissue.

Finally, to compare BPS/BPSG and BPA/BPAG disposition in the materno-fetal unit, a compartmental TK model described previously for BPA (Gauderat et al., 2017) was used to simulate the plasma concentration-time profiles of maternal and fetal free BPA and BPAG after repeated BPA maternal IV dosing under the same maternal exposure scenario as that used for BPS in experiment 2 (Fig. 5). Maximum maternal internal exposure (at the end of the infusion and for the last administration) to BPS and BPSG was around two times higher compared to the corresponding level of BPA and BPAG, due to the lower BPS and BPSG maternal clearances. However, the trough BPS and BPSG concentrations were of the same order of magnitude than the corresponding pattern observed for BPA and BPAG.

BPSG fetal exposure was much lower than that of BPAG (twenty times), reflecting the low placental materno-fetal transfer of BPS.

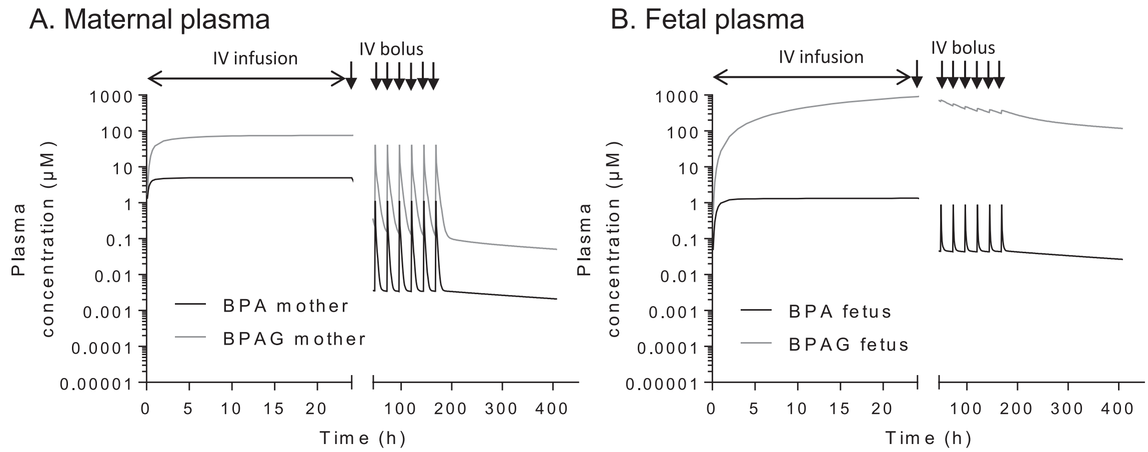

Fig. 5. Semilogarithmic plots of the simulated concentration-time-profiles (μM) for BPA (black line) and BPAG (grey line) in maternal (A) and fetal (B) plasma after an IV infusion loading dose of BPA (300 μmol/kg, 69.7 mg/kg) followed by the administration of repeated BPS IV bolus maintenance doses (20 μmol/kg, 4.65 mg/kg) to the mother. This dosing scheme was the same that was used for BPS in experiment 2 (see (Gauderat et al., 2017) for the toxicokinetic model).

Nonetheless, the BPS levels in fetal compartment was of similar magnitude to the corresponding levels of BPA, and the persistence of BPS and BPSG in the fetal compartment and of BPSG in the maternal compartment was similar to those of BPA/BPAG. Indeed, the elimination of these conjugated bisphenols from the fetal compartment depends on the hydrolysis rate and on the feto-maternal placental transfer of the back-converted conjugates. However, only 26% of the BPS dose steadily reentering the fetal circulation by this pathway was eliminated by placental transfer, compared with 70% for BPA. Thus, a large proportion of back-converted BPS can once again be reconjugated by the fetus, which would account for a long persistence of BPS, similar to that of BPA, in the fetal compartment.

5. Conclusions

This study provides new and critical information about the determinants of BPS fetal exposure and highlights differences with those prevailing for BPA. The limited materno-fetal placental transfer of BPS, ten times lower than for BPA (Gauderat et al., 2016) might lead to the hasty conclusion that the fetus is better protected from exposure to BPS than to BPA. But this is not the case. Indeed, by using our pregnant ewe model, we have demonstrated that the slow elimination of fetal BPSG due to its prior hydrolysis by the fetus and the limited feto-maternal placental passage of BPS, lead to prolonged fetal exposure both to BPS and to BPSG, which might facilitate deconjugation-reconjugation recycling. These mechanisms increased the fetal exposure to bioactive BPS by 87%. Ultimately, although weakly transferred across the placenta, BPS was able to accumulate in the fetal compartment. This means that under steady state conditions, after repeated gestational BPS/BPA exposure, the BPS in the fetal compartment can attain similar persistently levels to those of BPA.

Predictive models of human fetal exposure to BPS should now be carried out to put into perspective our genuine *in vivo* results regarding the pregnant woman. To do this, placental and fetal BPS metabolism differences during pregnancy and between species should now be investigated to determine the relevance of this animal model when evaluating the danger of fetal exposure to BPS in humans. The health implications of sustained fetal exposure to BPS and BPSG need to be addressed, especially in sensitive target tissue.

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Competing financial interests

The authors declare no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envint.2018.08.019.

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